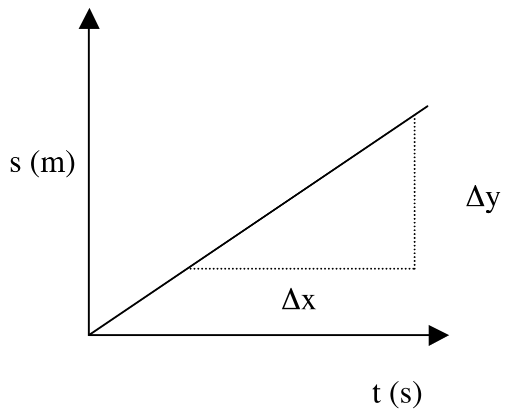
MOTION GRAPHS

DISPLACEMENT - TIME

CONSTANT VELOCITY

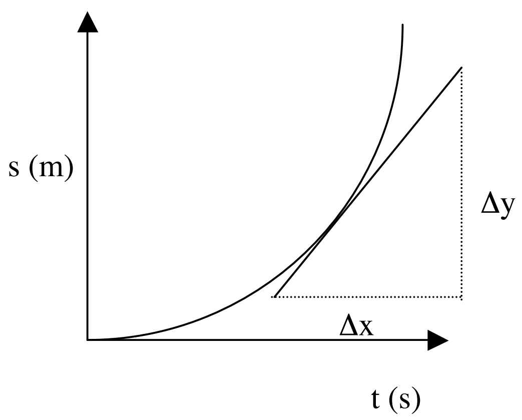


The slope or gradient of the line is given by:

$$\text{slope} = \frac{\Delta y}{\Delta x} = \frac{s}{t}$$

Hence the slope of the graph gives the **constant velocity** of the body or object.

CONSTANT ACCELERATION



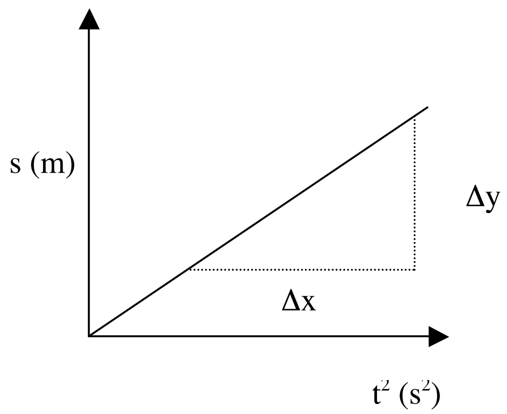
The graph at left represents an object falling under gravity or one that is uniformly accelerated across the ground.

The slope at a point on the curve gives the **instantaneous velocity** of the object at that time.

The graph is a parabola and clearly suggests that $s \propto t^2$.

The equation relating displacement and acceleration is $s = ut + 1/2\ at^2$.

If $s \propto t^2$, then plotting s versus t^2 should produce a straight line graph.



The slope or gradient represents 1/2 a.

The intercept represents ut.

The general equation of a line is:

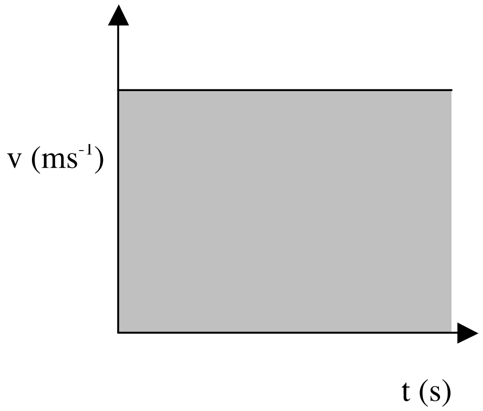
$$y = mx + b$$

b represents ut while m = 1/2 a.

IMPORTANT
The slope of a linear graph generally means something - it is related to an equation that is applicable. You must know what the slope or gradient represents!

VELOCITY - TIME

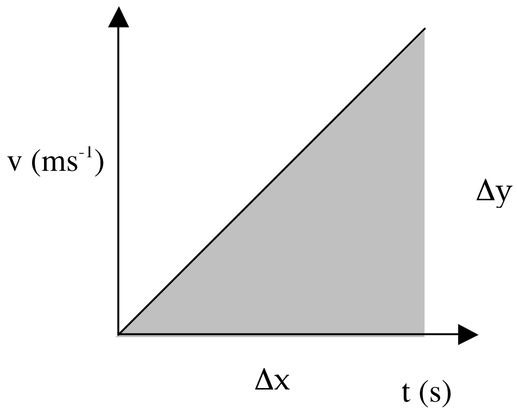
CONSTANT VELOCITY



The area under a v - t graph represents the displacement for the body.

i.e. $s = v \times t$

CONSTANT ACCELERATION



The slope gives the **acceleration** of the body.

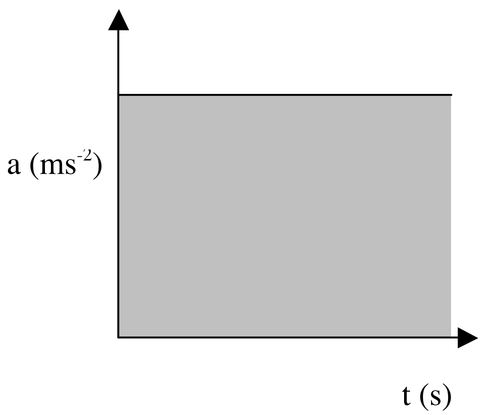
$$\text{slope} = \frac{\Delta v}{\Delta t} = a$$

Area under graph = displacement

= 1/2 x base x height

ACCELERATION - TIME

CONSTANT ACCELERATION



Area under graph = change in velocity

= Δv

EXAMPLE

A car travels at 20.0 ms^{-1} as it passes an observer on the side of the road. The car maintains this velocity for 4.00 s before decelerating uniformly for 3.00 s to 10.0 ms^{-1}. In the next 4.00 s it uniformly increases to 25.0 ms^{-1} before continuing at constant velocity. The entire motion occurs in a straight line.

(a) Draw a velocity - time graph for the first 11.0 s of the motion.
(b) Calculate the displacement of the car in this time.
(c) Draw an acceleration - time graph for the motion.